

Add a smoother.

Usage

```
stat_smooth(mapping = NULL, data = NULL, geom = "smooth", position = "identity",
  method = "auto", formula = y ~ x, se = TRUE, n = 80, fullrange = FALSE, level = 0.95,
  na.rm = FALSE, ...)
```

Arguments

method

smoothing method (function) to use, eg. `lm`, `glm`, `gam`, `loess`, `rlm`. For datasets with $n < 1000$ default is `loess`. For datasets with 1000 or more observations defaults to `gam`, see `gam` for more details.

formula

formula to use in smoothing function, eg. `y ~ x`, `y ~ poly(x, 2)`, `y ~ log(x)`

se

display confidence interval around smooth? (TRUE by default, see level to control

fullrange

should the fit span the full range of the plot, or just the data

level

level of confidence interval to use (0.95 by default)

n

number of points to evaluate smoother at

na.rm

If `FALSE` (the default), removes missing values with a warning. If `TRUE` silently removes missing values.

...

other arguments are passed to smoothing function

mapping

The aesthetic mapping, usually constructed with `aes` or `aes_string`. Only needs to be set at the layer level if you are overriding the plot defaults.

data

A layer specific dataset - only needed if you want to override the plot defaults.

geom

The geometric object to use display the data

position

The position adjustment to use for overlapping points on this layer

Value

a data.frame with additional columns `ypredicted` value `yminlower` pointwise confidence interval around the mean `yupper` pointwise confidence interval around the mean `sestandard` error

Description

Aids the eye in seeing patterns in the presence of overplotting.

Details

Calculation is performed by the (currently undocumented) `predictdf` generic function and its methods. For most methods the confidence bounds are computed using the `predict` method - the exceptions are `loess` which uses a t-based approximation, and for `glm` where the normal confidence interval is constructed on the link scale, and then back-transformed to the response scale.

Aesthetics

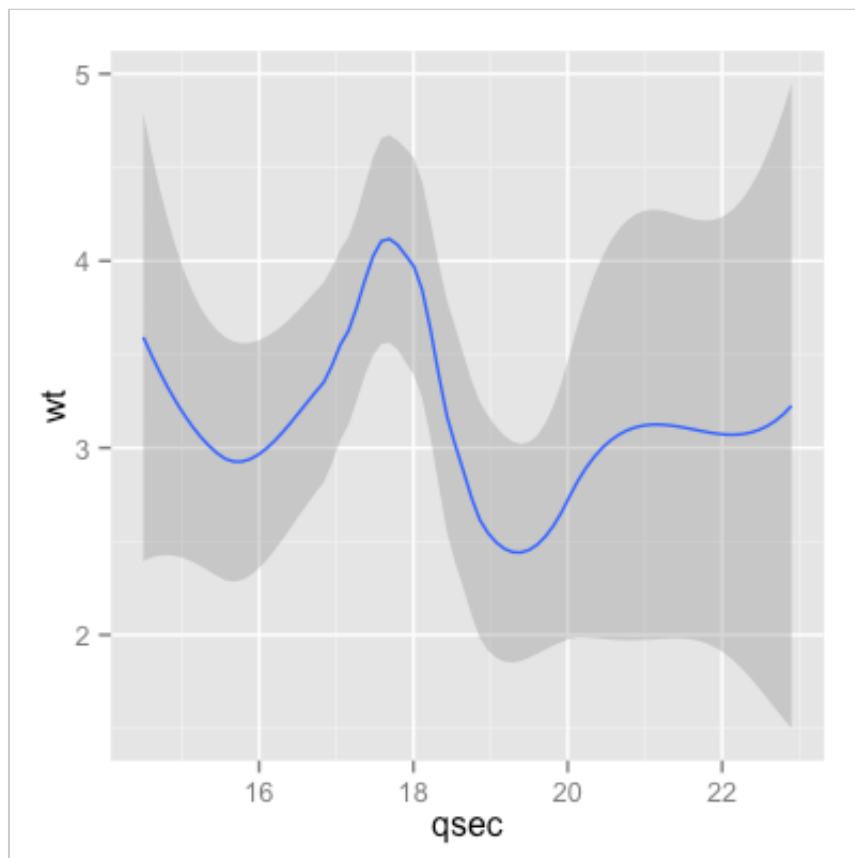
`stat_smooth` understands the following aesthetics (required aesthetics are in bold):

- **x**
- **y**

Examples

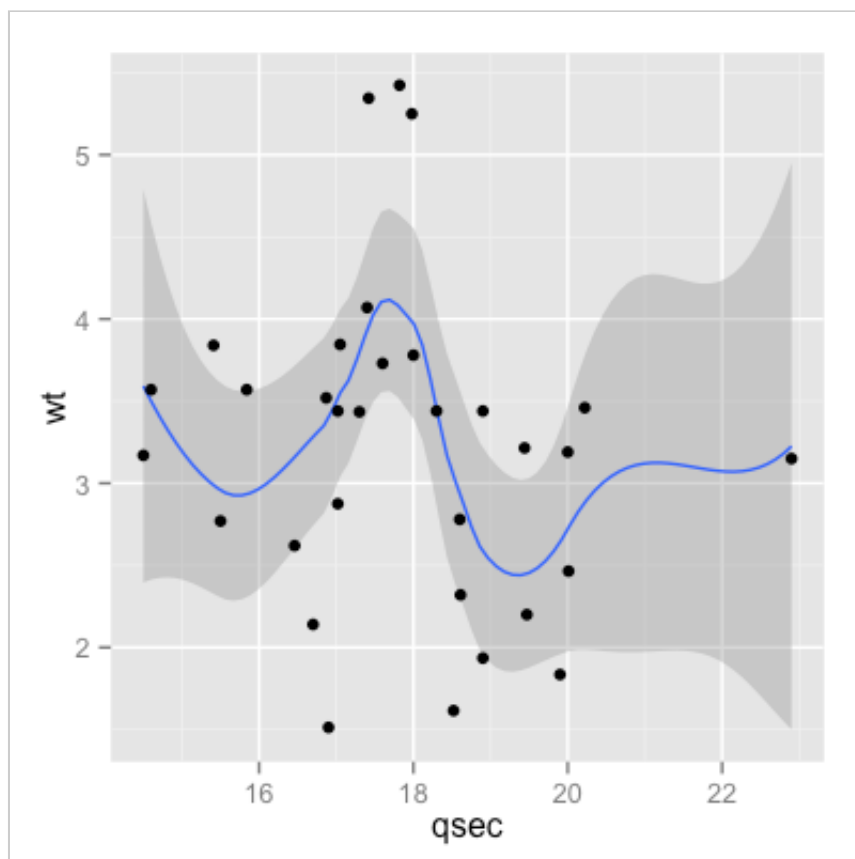
```
c <- ggplot(mtcars, aes(qsec, wt))  
c + stat_smooth()
```

`geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.`



```
c + stat_smooth() + geom_point()
```

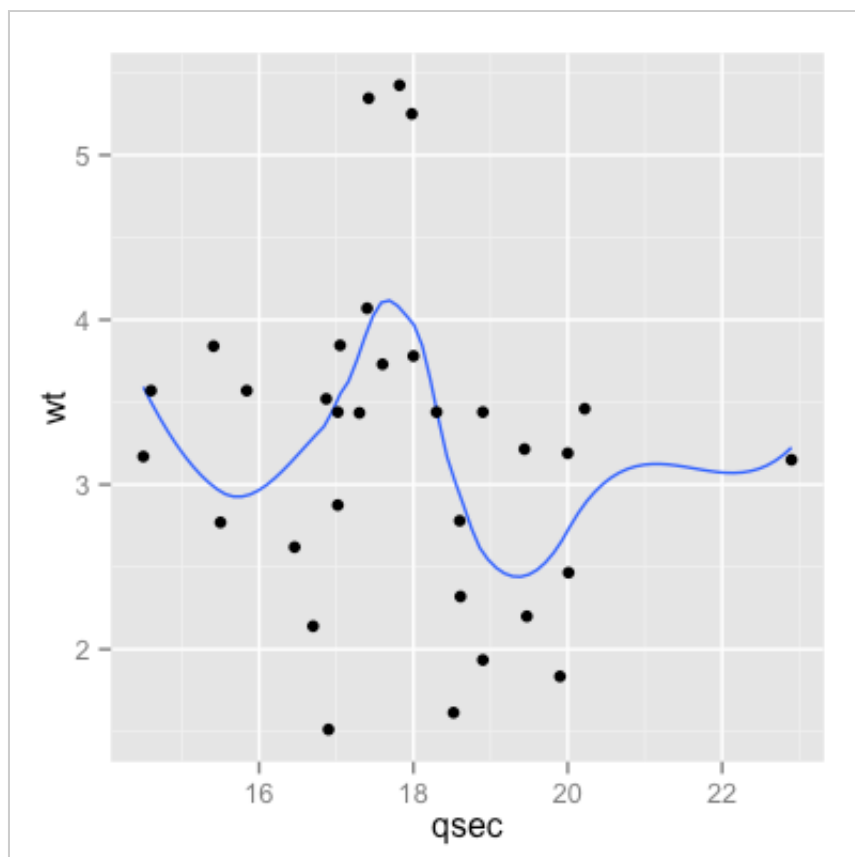
`geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.`



```
# Adjust parameters
```

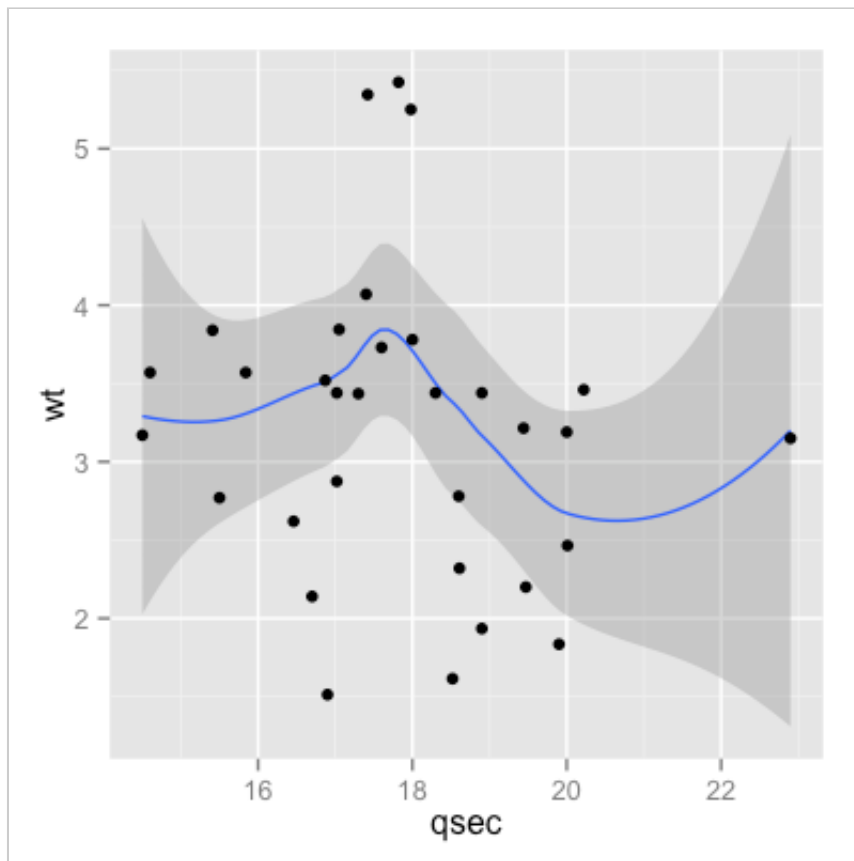
```
c + stat_smooth(se = FALSE) + geom_point()
```

geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.



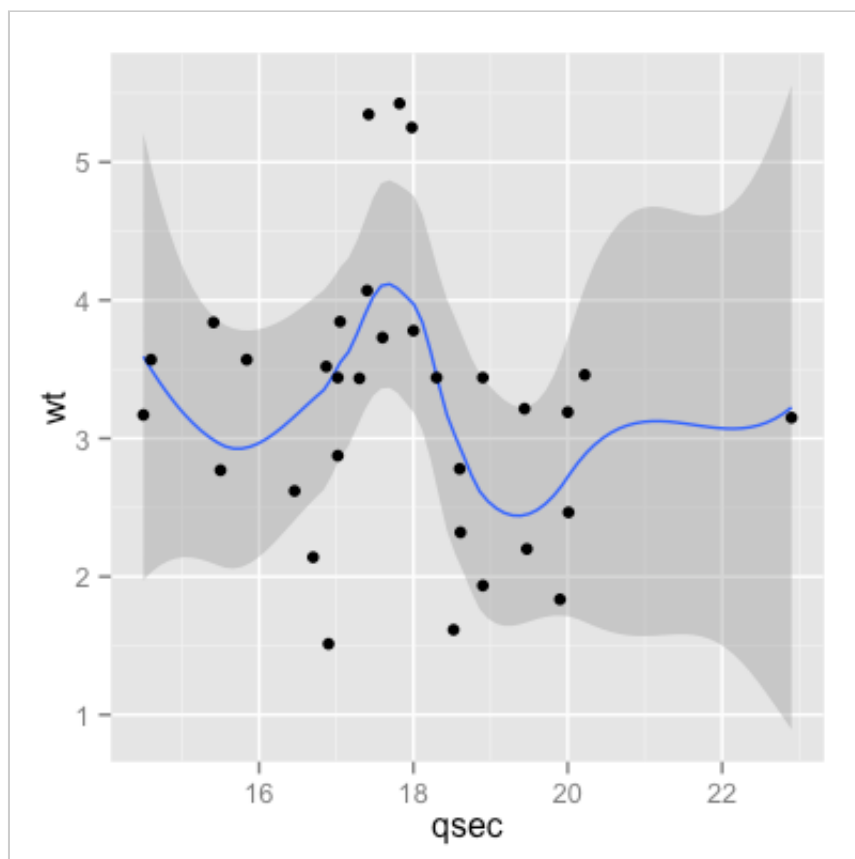
```
c + stat_smooth(span = 0.9) + geom_point()
```

geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.

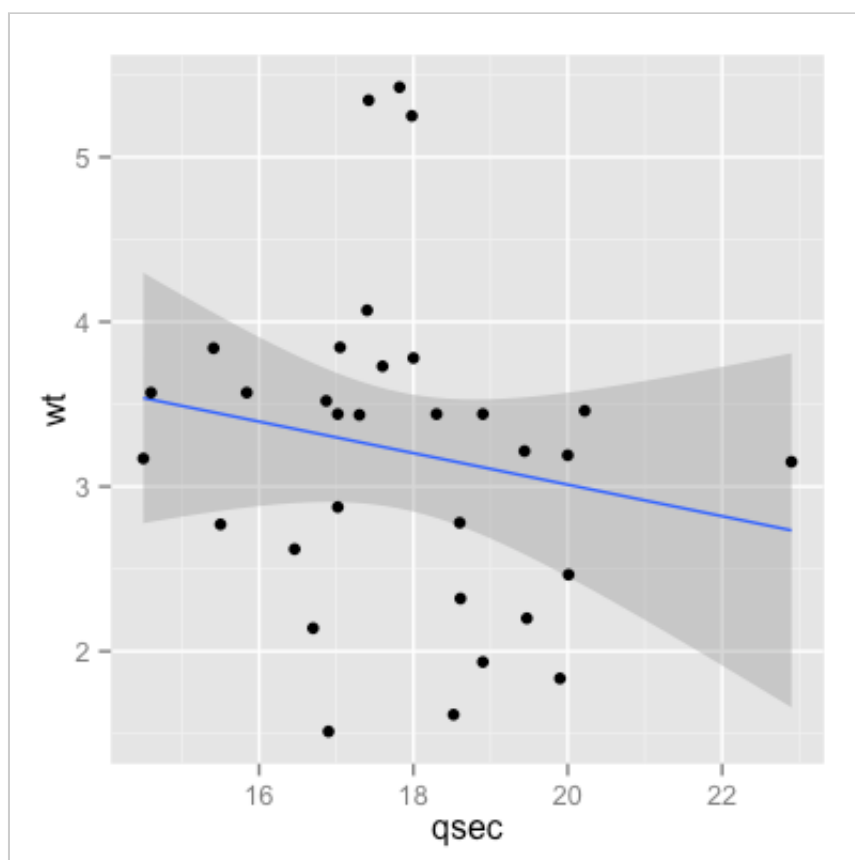


```
c + stat_smooth(level = 0.99) + geom_point()
```

geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.

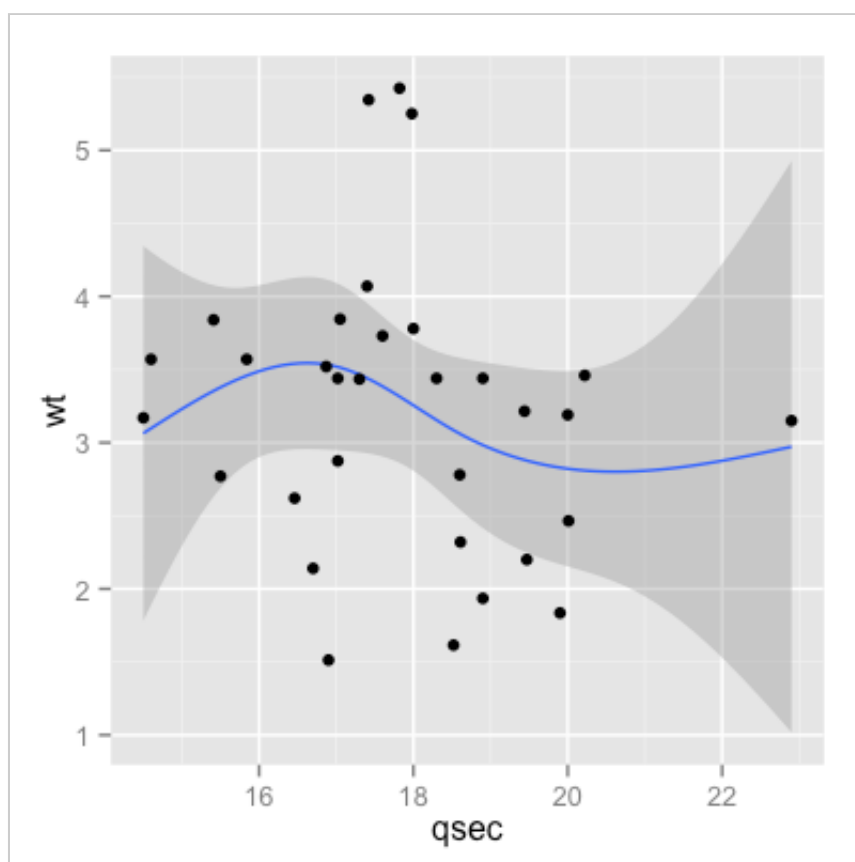


```
c + stat_smooth(method = "lm") + geom_point()
```

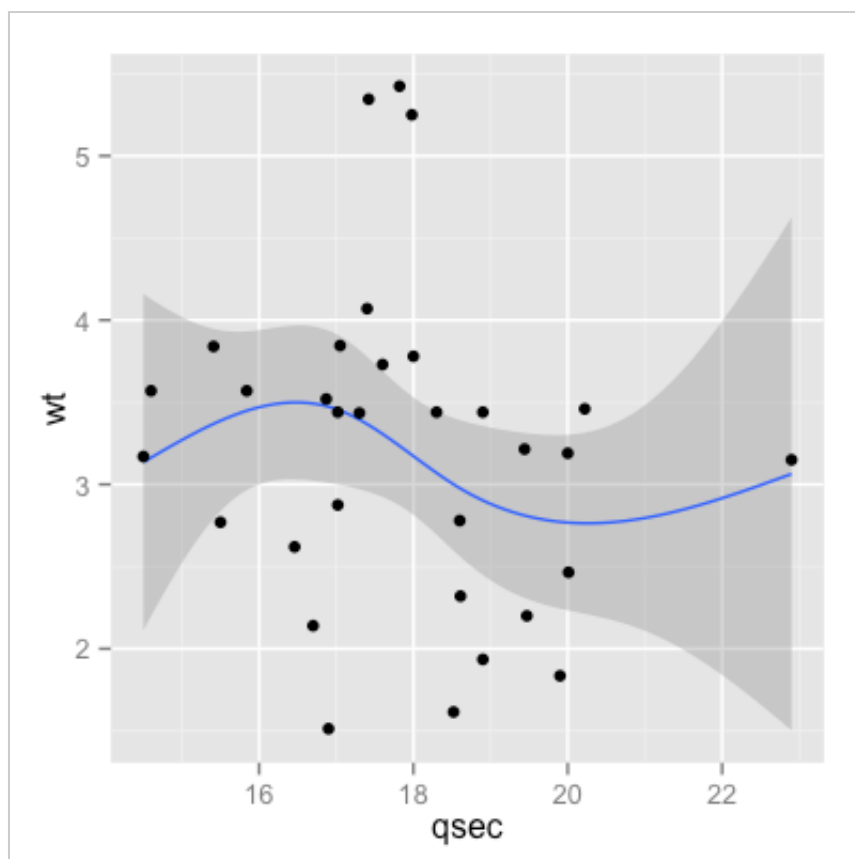


```
library(splines)
library(MASS)
c + stat_smooth(method = "lm", formula = y ~ ns(x,3)) +
```

```
geom_point()
```



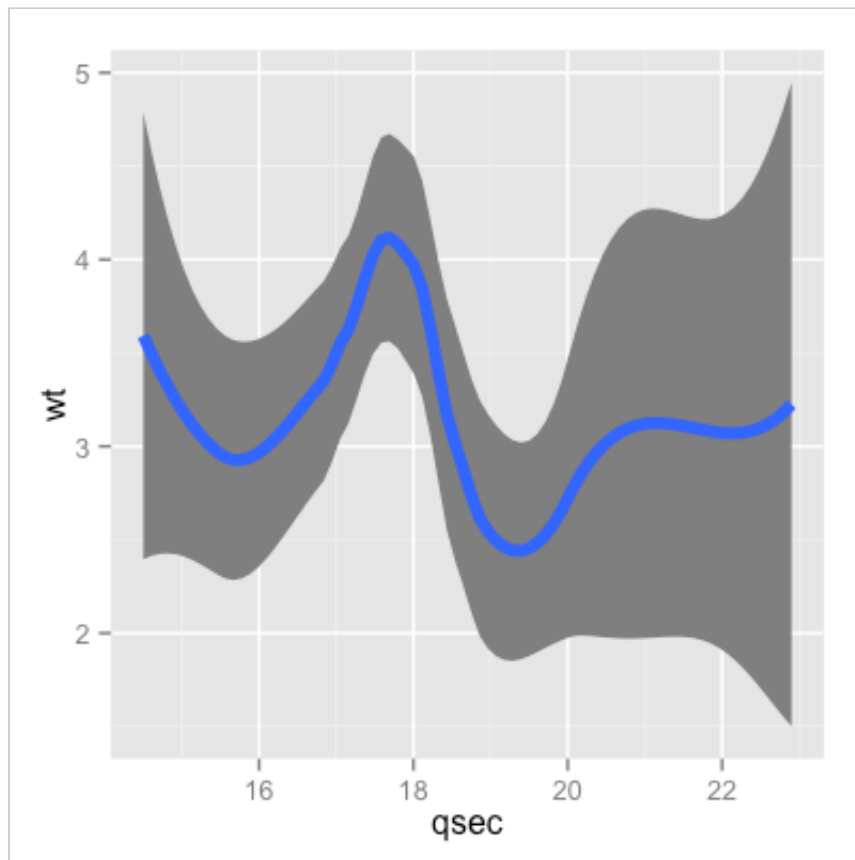
```
c + stat_smooth(method = rlm, formula= y ~ ns(x,3)) + geom_point()
```



```
# The default confidence band uses a transparent colour.
```

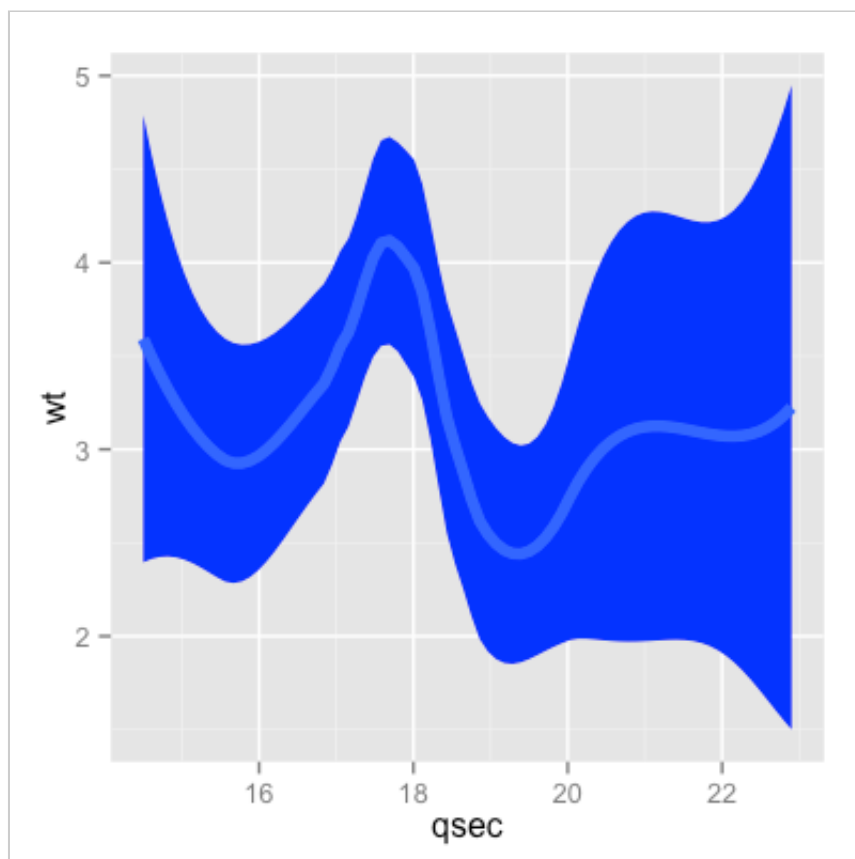
```
# This currently only works on a limited number of graphics devices  
# (including Quartz, PDF, and Cairo) so you may need to set the  
# fill colour to an opaque colour, as shown below  
c + stat_smooth(fill = "grey50", size = 2, alpha = 1)
```

geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.



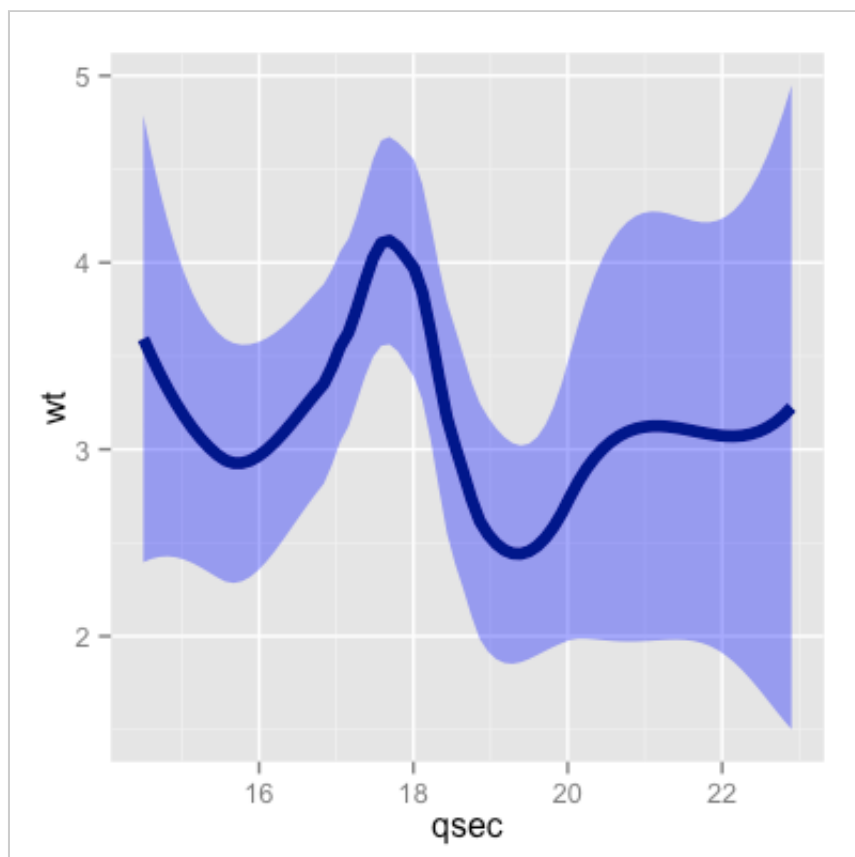
```
c + stat_smooth(fill = "blue", size = 2, alpha = 1)
```

geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.



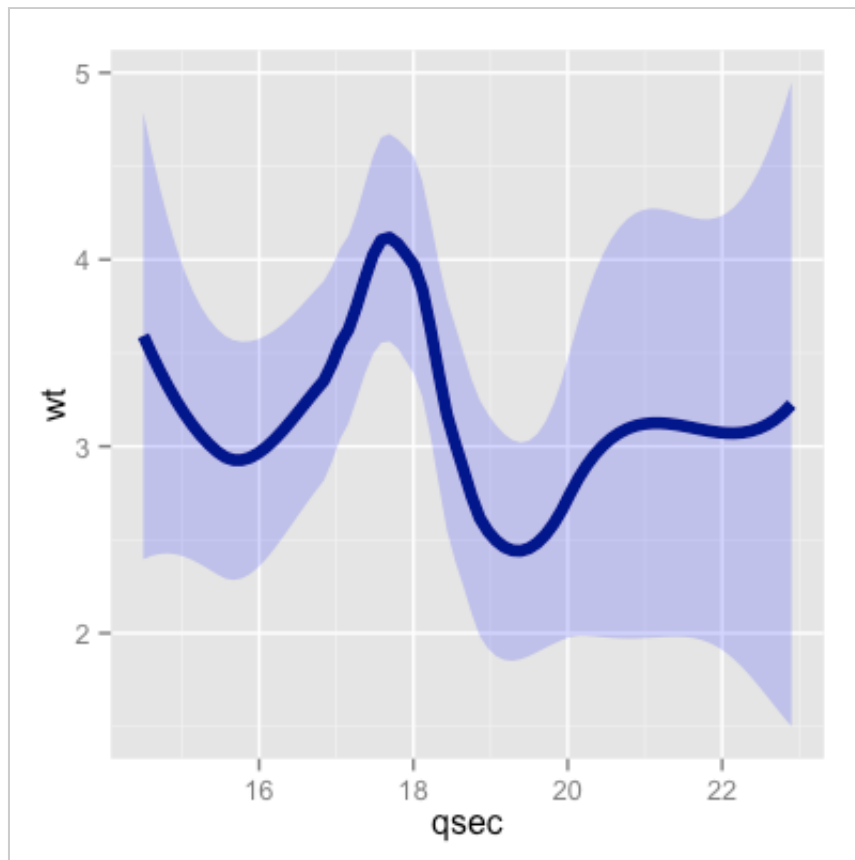
```
# The colour of the line can be controlled with the colour aesthetic  
c + stat_smooth(fill="blue", colour="darkblue", size=2)
```

geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.



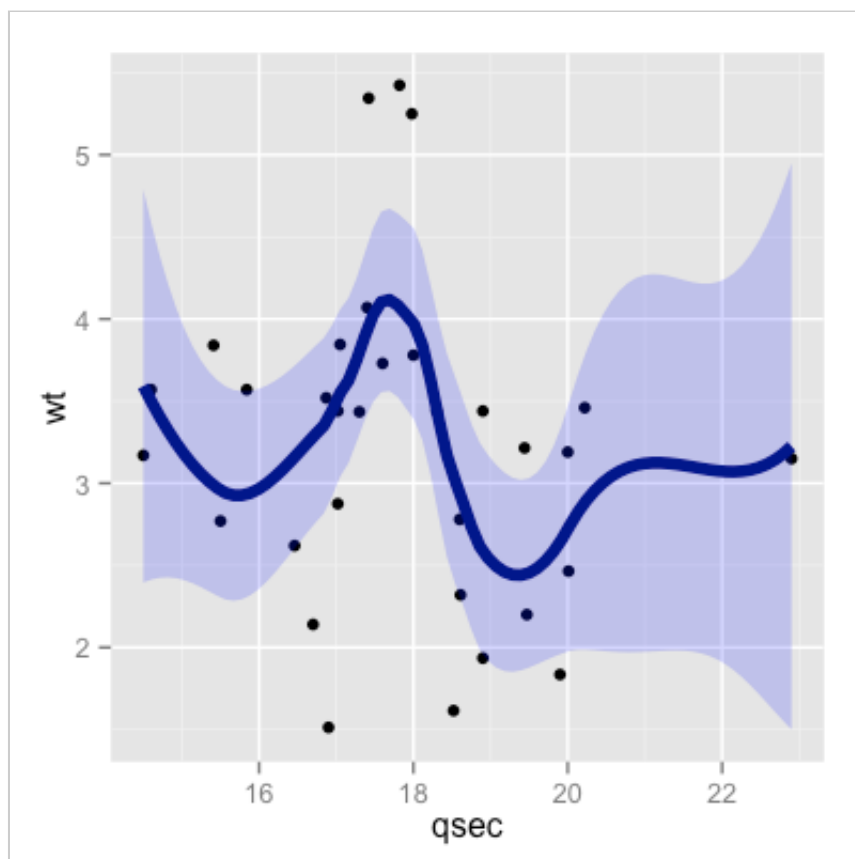

```
c + stat_smooth(fill="blue", colour="darkblue", size=2, alpha = 0.2)
```

geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.

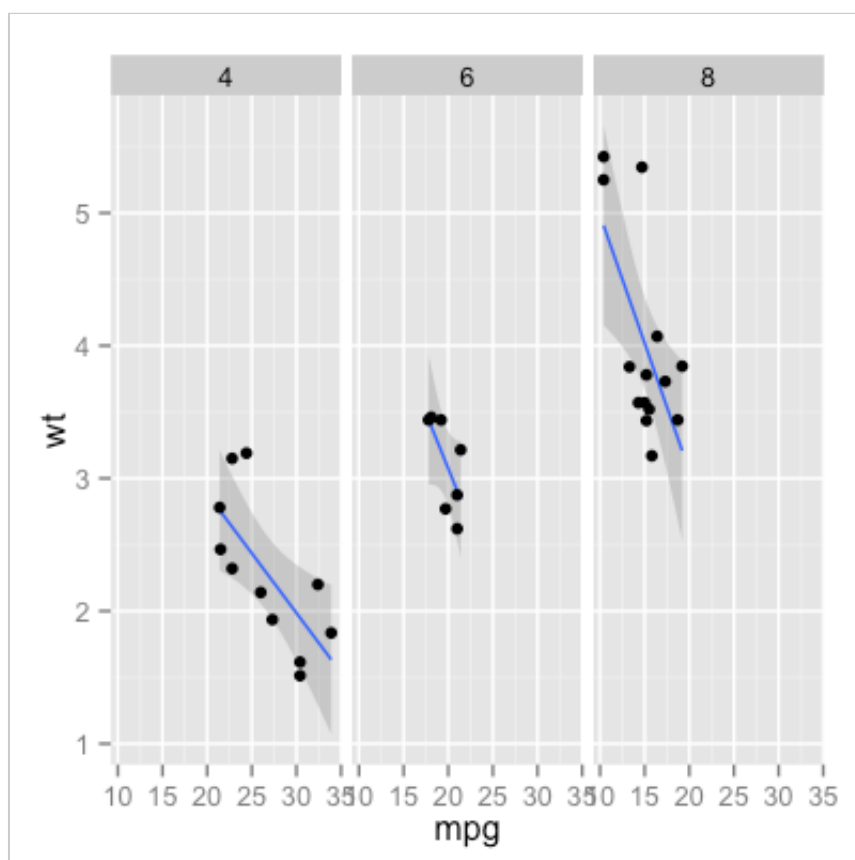


```
c + geom_point() +  
  stat_smooth(fill="blue", colour="darkblue", size=2, alpha = 0.2)
```

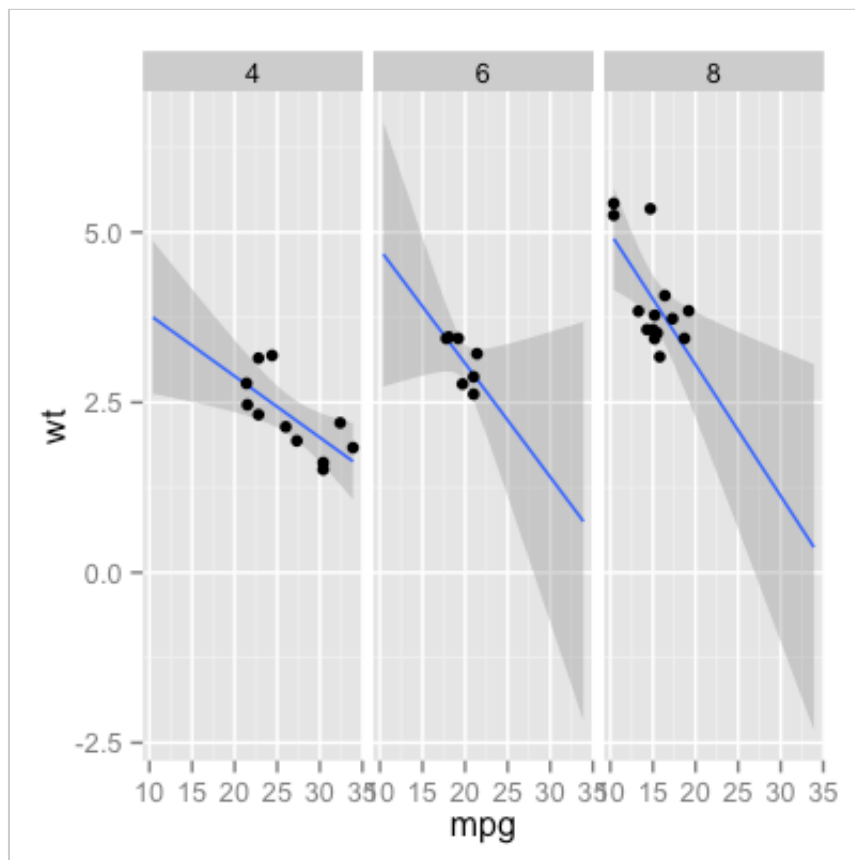
geom_smooth: method="auto" and size of largest group is <1000, so using loess. Use 'method = x' to change the smoothing method.



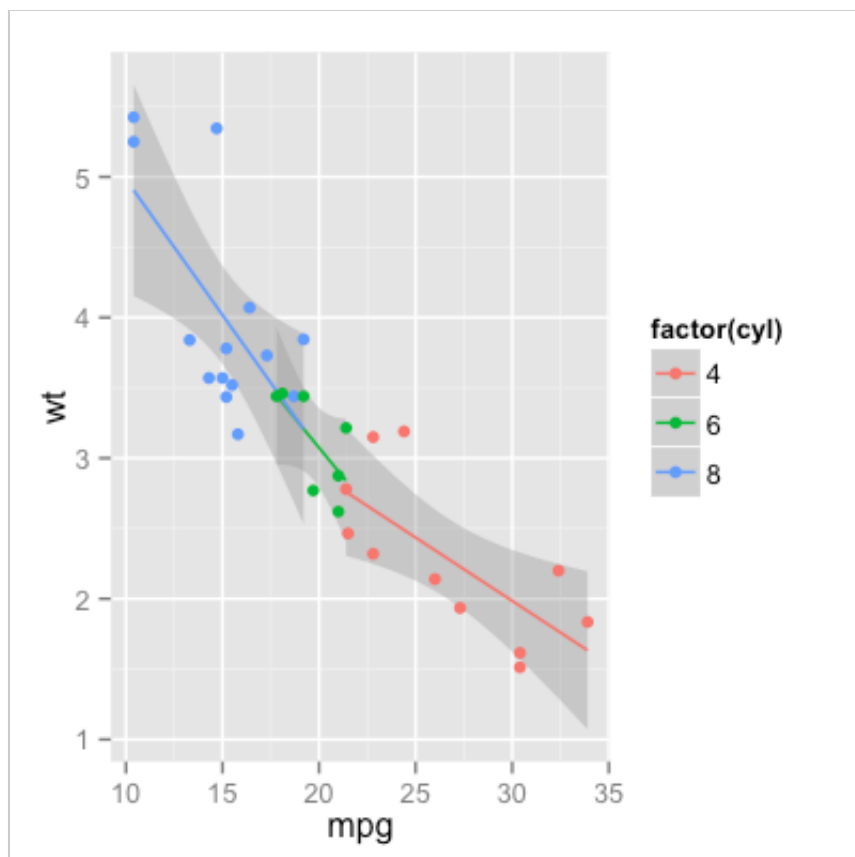
```
# Smoothers for subsets
c <- ggplot(mtcars, aes(y=wt, x=mpg)) + facet_grid(. ~ cyl)
c + stat_smooth(method=lm) + geom_point()
```



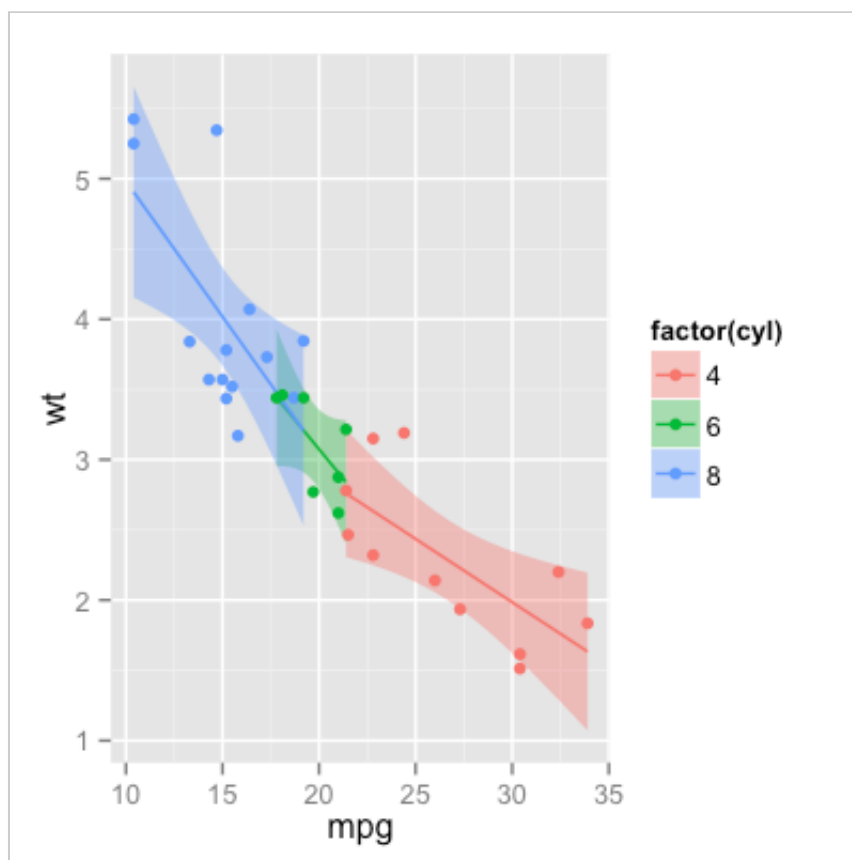
```
c + stat_smooth(method=lm, fullrange = TRUE) + geom_point()
```



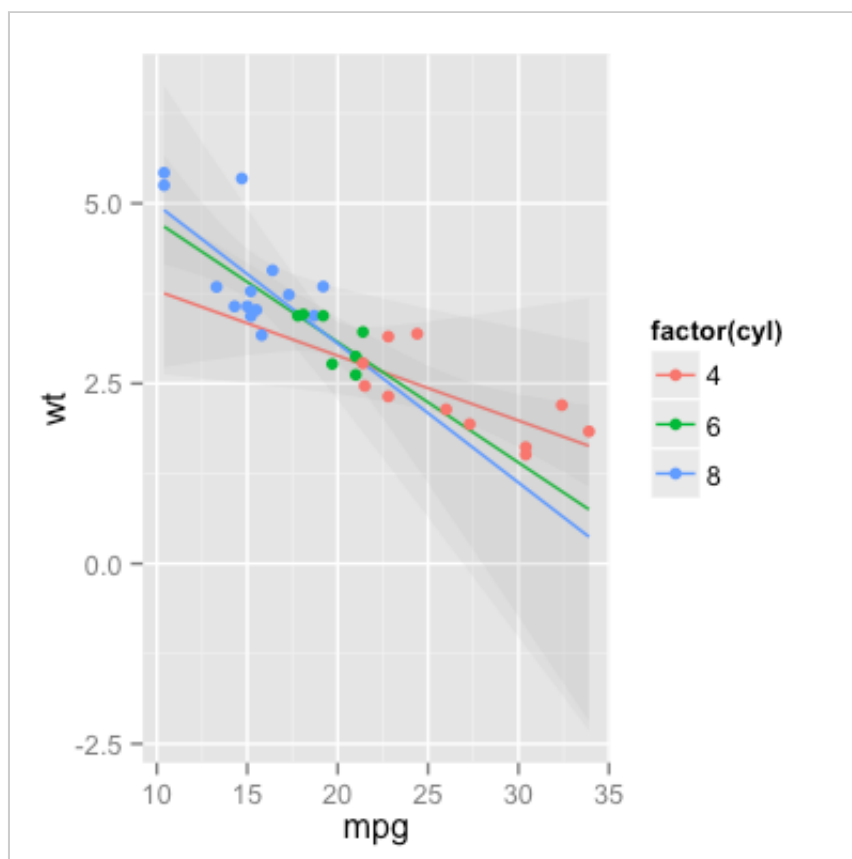
```
# Geoms and stats are automatically split by aesthetics that are factors
c <- ggplot(mtcars, aes(y=wt, x=mpg, colour=factor(cyl)))
c + stat_smooth(method=lm) + geom_point()
```



```
c + stat_smooth(method=lm, aes(fill = factor(cyl))) + geom_point()
```



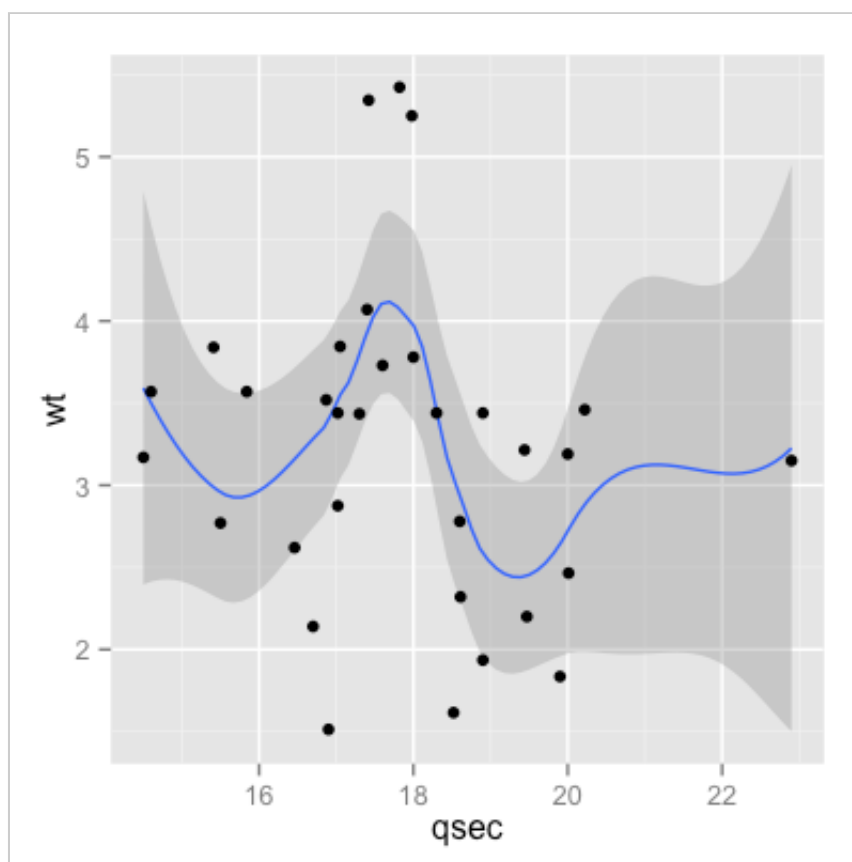
```
c + stat_smooth(method=lm, fullrange=TRUE, alpha = 0.1) + geom_point()
```



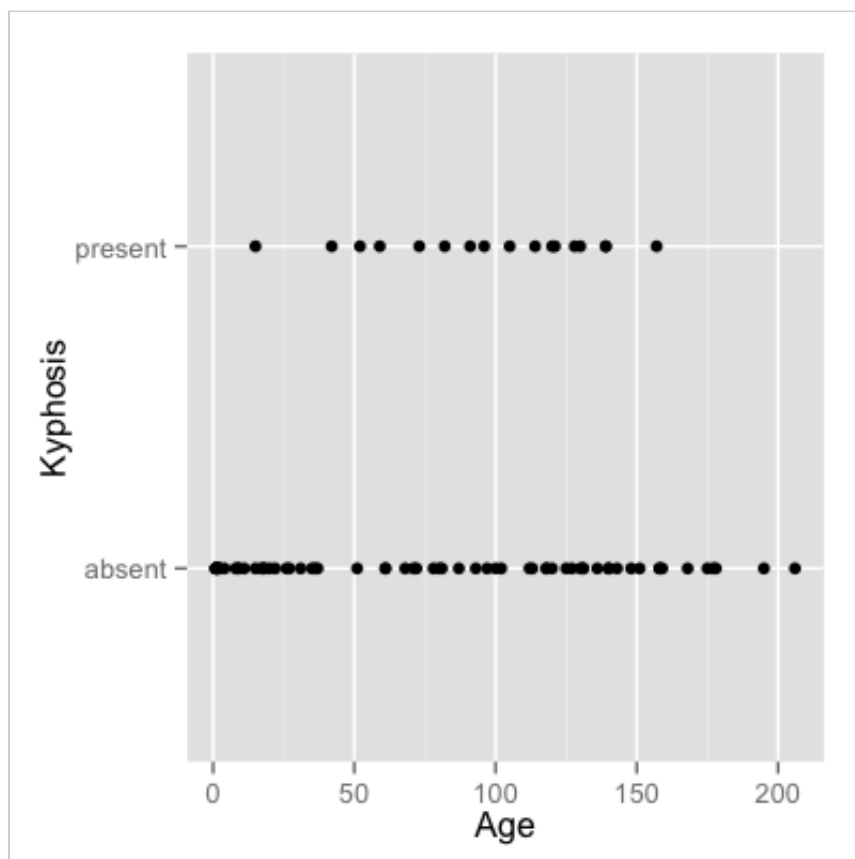
```
# Use qplot instead
```

```
qplot(qsec, wt, data=mtcars, geom=c("smooth", "point"))
```

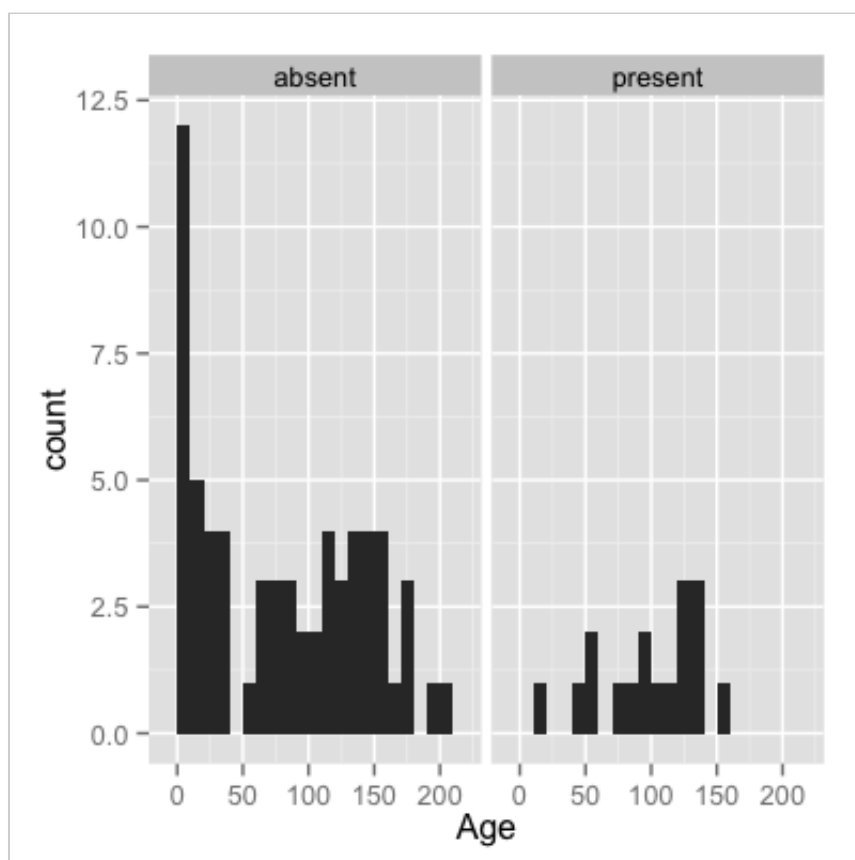
`geom_smooth`: `method="auto"` and size of largest group is `<1000`, so using `loess`. Use `'method = x'` to change the smoothing method.



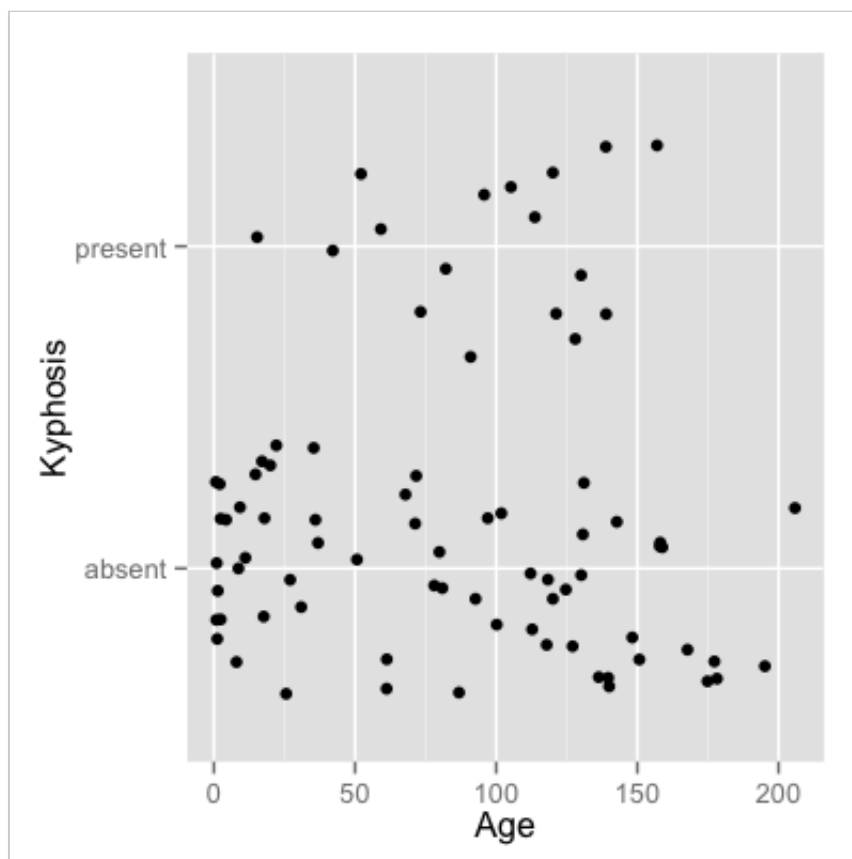
```
# Example with logistic regression  
data("kyphosis", package="rpart")  
qplot(Age, Kyphosis, data=kyphosis)
```



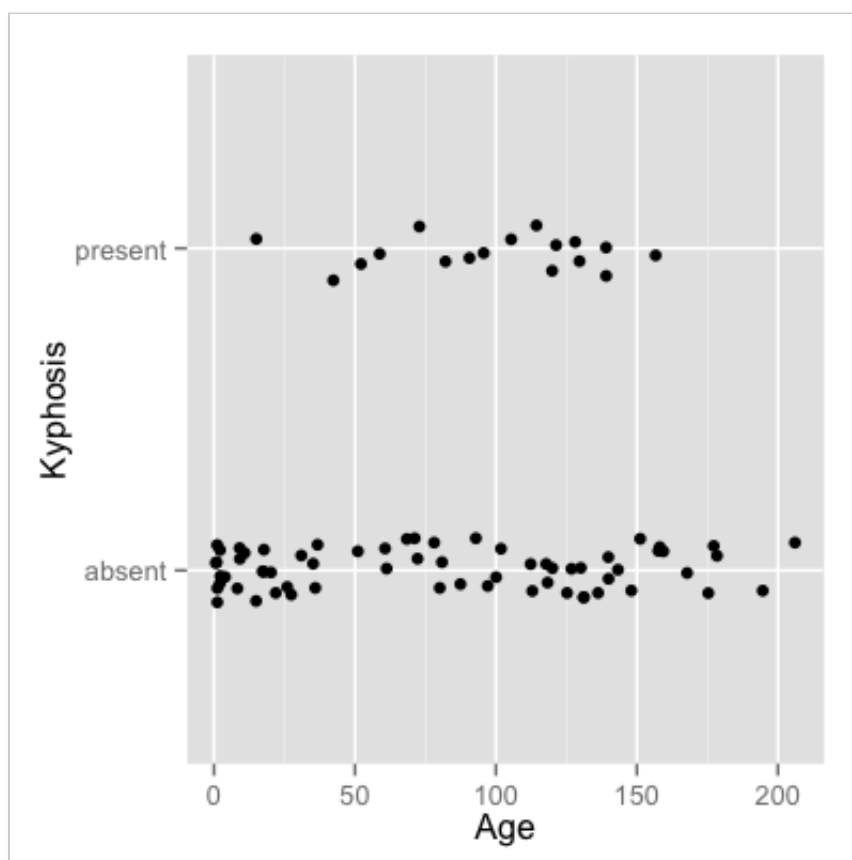
```
qplot(Age, data=kyphosis, facets = . ~ Kyphosis, binwidth = 10)
```



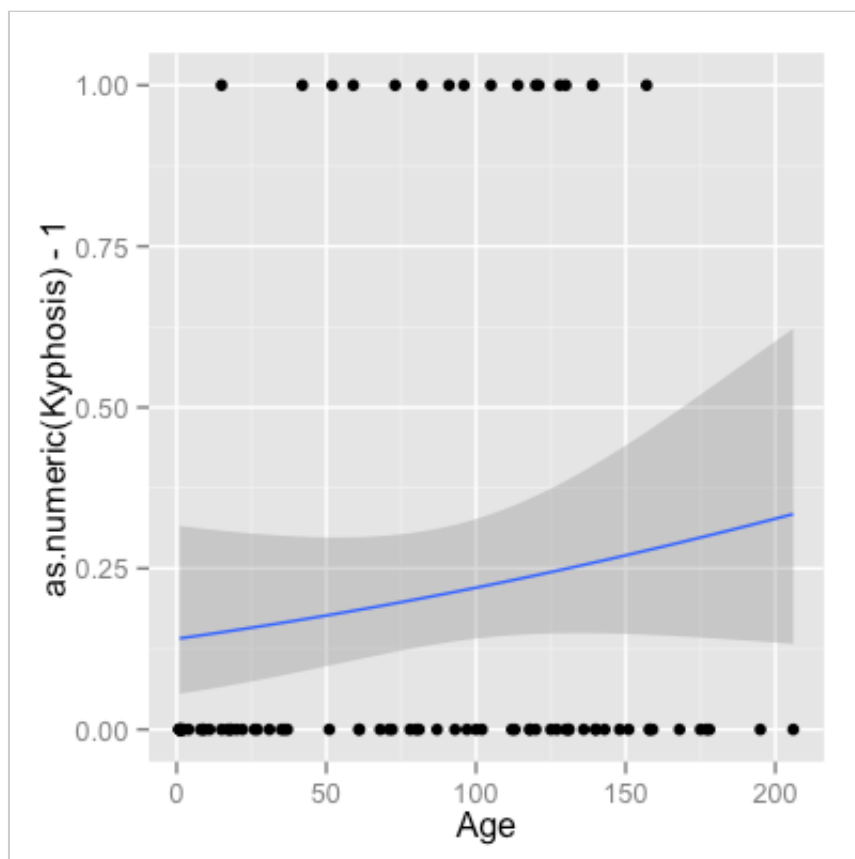
```
qplot(Age, Kyphosis, data=kyphosis, position="jitter")
```



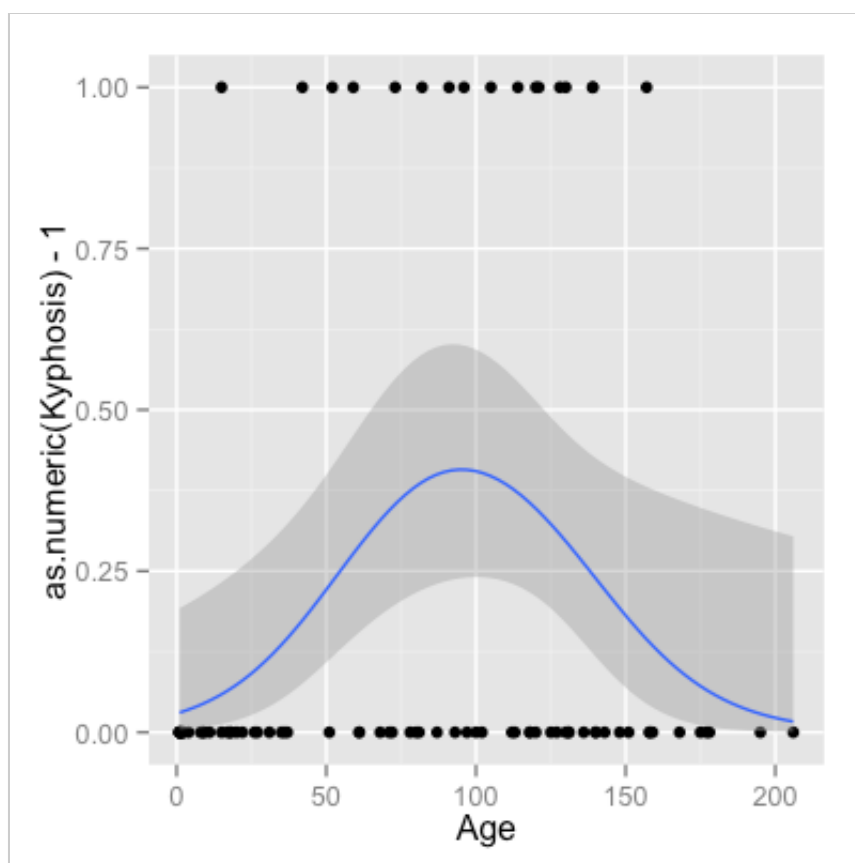
```
qplot(Age, Kyphosis, data=kyphosis, position=position_jitter(height=0.1))
```



```
qplot(Age, as.numeric(Kyphosis) - 1, data = kyphosis) +  
  stat_smooth(method="glm", family="binomial")
```



```
qplot(Age, as.numeric(Kyphosis) - 1, data=kyphosis) +
  stat_smooth(method="glm", family="binomial", formula = y ~ ns(x, 2))
```



See also

`lm` for linear smooths, `glm` for generalised linear smooths, `loess` for local smooths

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